



ML Pipeline Sustainability Impact Report

Company: MetaVerdax

Report Period: 2026-04-17 to 2026-04-17

Generated: 2026-04-17 08:17 UTC

Scope: 2026-04-17 to 2026-04-17 (inclusive, by date(timestamp))

Executive Summary

During the period 2026-04-17 to 2026-04-17, Verdax monitored **0 training pipeline runs** for MetaVerdax. Of these, **0 retrains were prevented** due to detected data anomalies, saving an estimated **0.0 kg of CO₂ emissions** and **\$0.00 in compute costs**.

Impact Metrics

All metrics below are date-scoped to 2026-04-17 through 2026-04-17 and derived from logged incidents/carbon records.

Metric	Value
Total Pipeline Runs	0
Retrains Prevented	0
CO ₂ Saved	0.0 kg
Energy Saved	0.0 kWh
Cost Saved	\$0.00
Equivalent Flights Avoided	0.00
Equivalent Car km Avoided	0

Reporting Scope & Provenance

All totals are scoped by date(timestamp) using inclusive start/end filters. Carbon totals come only from prevented retrains recorded in the Verdax database ledger.

Incident rows in scope: 0; blocked rows: 0; carbon records: 0.

CO₂ Saved Over Time

No carbon savings data available for this period.

Incidents Over Time

No incidents recorded for this period.

Incident Log

No incidents recorded for this period.

Methodology & Sources

Verdax reports carbon impact using published training-energy benchmarks, regional grid-intensity factors, and explicit conversion formulas.

1. Carbon Calculation Framework

Formula: **Energy (kWh) × Grid Intensity (kg CO₂/kWh) = CO₂ (kg)**

Model size	Base energy (kWh)	Benchmark source
Small	10	Patterson et al. 2021 / conservative small-model benchmark
Medium	100	Patterson et al. 2021 / BERT-scale benchmark
Large	1,000	Patterson et al. 2021 / large-model benchmark
LLM	1,287,000	Luccioni et al. 2023 / BLOOM training emissions

2. Grid Intensity Factors

Region	kg CO ₂ /kWh	Source	Last updated
us-east-1	0.386	EPA eGRID 2022	Apr 2026
eu-west-1	0.233	EPA eGRID 2022 / EU proxy	Apr 2026
ap-southeast-1	0.493	IEA regional average	Apr 2026
on_prem	0.475	Conservative global average	Apr 2026

3. Equivalency Conversions

1 transatlantic flight ≈ 1,000 kg CO₂ (ICAO Carbon Calculator).

1 km driving ≈ 0.21 kg CO₂ (EPA average passenger vehicle).

4. Limitations & Disclaimer

Estimates are order-of-magnitude benchmarks. Actual emissions vary by hardware, utilization, data-center PUE, and workload parallelism. Verdax uses conservative published averages rather than live cloud billing telemetry.

Methodology version: 1.0 — April 2026. Next review: October 2026.

